

RESEARCH

Open Access



Effect of Vitamin C on bleeding and pain after kidney biopsy: a randomized controlled trial

Jipeng Li^{1†}, Fang Zhao^{1,2†}, Rui Lu^{1†}, Yang Zha¹, Fengpei Chang¹, Xiayin Li¹, Jingli Gao¹, Xi Shen¹, Qing Jia¹, Jing Zhao¹, Xiaolan Che¹, Xiaomeng Xu¹, Mengting Wu¹, Hongyi Zhang¹, Yuting Zhang¹, Ye Chen¹, Xinrui Tan^{1,2} and Lijie He^{1*}

Abstract

Background Percutaneous renal biopsy (PRB) is an essential diagnostic procedure for kidney diseases, yet postbiopsy perirenal hematoma and pain remain frequent complications. Vitamin C has emerging evidence supporting its role in reducing perioperative bleeding and pain, but its efficacy in the context of PRB is unknown.

Methods This single-center, prospective, open-label, randomized controlled trial enrolled 710 adults with chronic kidney disease (CKD) scheduled for an ultrasound-guided native kidney biopsy. Participants were randomly assigned (1:1) to receive perioperative intravenous vitamin C (20 mg/kg at 6 h before and 18 h after biopsy) or standard care alone. The primary outcomes were the area of perirenal hematoma on renal ultrasound at 24 h and self-reported pain intensity (Visual Analogue Scale, VAS) at 2, 12, and 24 h post-biopsy.

Results All 710 randomized participants were included in the analysis. The adjusted mean difference in hematoma area at 24 h was 4.44 mm² (95% CI, -52.31 to 61.18; P = 0.878). After multiple testing correction, there were no significant differences in the primary pain outcome (VAS scores) between groups at any time point; however, an exploratory analysis of pain severity categories indicated a less favorable distribution in the Vitamin C group at 12 h (P for trend = 0.033). The incidence of any perirenal hematoma was 56.3% in the vitamin C group versus 50.1% in the control group (risk ratio, 1.12; 95% CI, 0.98–1.29; P = 0.098). Rates of other bleeding complications and adverse events were low and comparable between groups.

Conclusions In CKD patients undergoing PRB, perioperative intravenous vitamin C did not reduce the size of perirenal hematoma, alleviate postbiopsy pain, or decrease the incidence of bleeding complications within 24 h. These findings do not support the routine use of vitamin C for preventing complications after kidney biopsy.

Keywords Vitamin C, Percutaneous renal biopsy, Randomized controlled trial, Hematoma, Post-procedural pain

[†]Jipeng Li, Fang Zhao and Rui Lu contributed equally to this work.

*Correspondence:

Lijie He

43218204@qq.com; helijie@fmmu.edu.cn

¹Department of Nephrology, Xijing Hospital, Fourth Military Medical University, Xi'an, Shaanxi, China

²The Graduate School, Xi'an Medical University, Xi'an, Shaanxi, China



© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

Background

Percutaneous renal biopsy (PRB) is an indispensable diagnostic procedure for a wide spectrum of kidney diseases. While generally safe under ultrasound guidance, it carries an inherent risk of procedure-related complications, most notably perirenal hematoma and localized pain [1, 2]. Post-biopsy pain primarily stems from direct mechanical injury to the local tissues or irritation caused by perirenal bleeding. Although most perirenal hematomas are small and the associated discomfort is transient and self-limiting, severe hematomas can lead to significant pain, anemia, hemodynamic instability, and in rare instances, necessitate invasive interventions such as transfusion or angioembolization [3, 4]. Despite advances in technique and imaging, there remains a need for safe, effective, and accessible pharmacological strategies to mitigate these risks, with a particular focus on preventing significant bleeding, which is a fundamental contributor to post-procedural morbidity.

The search for an optimal prophylactic agent has yielded limited and inconsistent evidence. Although desmopressin may reduce postbiopsy hematoma incidence [5], concerns remain regarding its safety profile, including a lack of clear postoperative bleeding benefit [6], associations with thrombotic events such as infarction [7], and the risk of provoking severe hyponatremia [8]. Conversely, the antifibrinolytic agent tranexamic acid has, in some studies, unexpectedly been associated with increased hematoma size [9], highlighting the complexity of modulating hemostasis in this setting. This gap underscores the necessity for alternative therapeutic pathways that are both effective and inherently safe.

Vitamin C or ascorbic acid has emerged as a promising candidate for mitigating post-biopsy complications, supported by a compelling biological rationale that extends to hemostasis and analgesia. On one hand, it promotes hemostasis by enhancing collagen synthesis in vessel walls [10] and protecting platelet function [11]. On the other hand, it may also help alleviate pain through its potential antioxidant and anti-inflammatory properties [12], which could mitigate tissue irritation and nociceptive signaling at the puncture site. Thus, with its potential to both reduce bleeding and alleviate pain, vitamin C may offer a broader therapeutic benefit compared to agents targeting hemostasis alone.

Previous clinical evidence have provided preliminary support. For instance, intravenous vitamin C has been associated with decreased bleeding events in gynecological and endoscopic surgeries [13, 14]. Similarly, several randomized trials and meta-analyses suggest that perioperative vitamin C supplementation can lower early postoperative pain scores [15, 16]. However, these findings remain heterogeneous and may be influenced by factors such as surgical setting, dosage, and administration

timing. More importantly, no study to date has specifically evaluated whether vitamin C can reduce procedure-related bleeding and pain following PRB.

Therefore, we designed a prospective randomized controlled trial to evaluate the efficacy and safety of perioperative intravenous vitamin C in participants undergoing PRB. The primary objectives are to assess its ability to reduce the incidence and severity of bleeding complications and to alleviate biopsy-related pain. This investigation aims to provide a novel, safe, and cost-effective adjuvant strategy to enhance recovery after PRB. The findings may provide high-quality evidence for the clinical feasibility of perioperative vitamin C use in such minimally invasive procedures.

Methods

Study conduct

This study was conducted in the Department of Nephrology at a large tertiary teaching hospital in China, and strictly adhered to the principles of the Declaration of Helsinki and relevant laws and regulations in China. The research protocol was approved by the institutional ethics committee and followed its stipulated requirements.

Design and participants

This was a single-center, prospective, open-label, randomized controlled trial with two parallel groups (allocation ratio 1:1). This study was in full compliance with the principles of the Declaration of Helsinki. The protocol is registered at the Chinese Clinical Trial Registry (ChiCTR2500106962).

The study enrolled adults with chronic kidney disease (CKD) who were scheduled for a percutaneous ultrasound-guided native kidney biopsy at the study site. Key exclusion criteria included a known hypersensitivity to vitamin C, active bleeding disorders, uncontrolled severe hypertension (systolic BP > 180 mmHg), structural kidney abnormalities, pregnancy, lactation, or oliguria. Full inclusion and exclusion criteria are provided in the protocol (Additional file 2).

Interventions

Eligible participants were randomized to either the intervention or control group. The intervention group received intravenous vitamin C (20 mg/kg, diluted in 100 mL of 0.9% sodium chloride) administered twice: at 6 h before and 18 h after the biopsy. The control group received standard care without the vitamin C infusion. The vitamin C dose was selected based on pharmacokinetic studies [17] and previous perioperative trials [13, 14].

Under ultrasound guidance, renal lower pole biopsy was conducted in the lateral decubitus position with local anesthesia using 5 mL of 2% lidocaine and a 16 G

Tru-Cut needle. After sterile gauze dressing and abdominal binder compression, patients remained supine for 8 h postoperatively. All procedures were performed by a single experienced senior specialist and her team to ensure operational standardization, based on practice guidelines [18] and local post-procedure protocol.

Outcomes

Primary Outcomes included: 1) The area of perirenal hematoma assessed by renal ultrasound at 24 h post-biopsy. 2) Pain intensity measured using a self-reported 100 mm visual analogue scale (VAS) at 2, 12, and 24 h.

Secondary Outcomes included the incidence and severity distribution of perirenal hematoma, the incidence of other bleeding events (gross hematuria, transfusion requirement or intervention for bleeding), the distribution of pain severity categories, and the incidence of adverse events (AEs).

Sample size

The sample size was calculated based on preliminary data from a previous trial (See Protocol). To detect a difference in hematoma area with 80% power at a two-sided alpha of 0.05, and accounting for a potential 20% reduction in effect size, a total of 710 participants (355 per group) were required.

Randomization and allocation

A computer-generated random sequence was created by an independent statistician using SPSS software. Participants were assigned a sequential study number upon enrollment, which corresponded to their pre-determined group allocation.

Blinding

This trial was conducted open-label. Participants and treating clinicians were aware of the treatment assignment. However, the primary outcome assessors (radiologists measuring hematoma area on ultrasound) were blinded to group allocation.

Statistical methods

Analyses followed the intention-to-treat (ITT) principle. Continuous variables were compared using t-tests or Mann-Whitney U tests. Categorical variables were compared using chi-square or Fisher's exact tests. The Cochran-Mantel-Haenszel (CMH) test was used to analyze trends in ordered categorical outcomes. For the co-primary pain outcome, VAS scores were analyzed both as continuous variables with Bonferroni correction for multiple time points and as ordinal categories. An analysis of covariance (ANCOVA) and a zero-inflated negative binomial regression model was pre-specified as a sensitivity analysis for hematoma area and VAS scores to adjust

the baseline confounder of age, respectively. A post-hoc exploratory subgroup analysis was performed based on baseline urinary vitamin C levels (tertiles and quartiles). The statistical methods are detailed in Supplementary Methods (Additional file 1). A two-sided P-value < 0.05 was considered significant. Analyses were performed using SPSS or R software.

Results

Patient enrollment and intervention implementation

Between August and December 2025, a total of 847 participants were consecutively screened for eligibility. Of these, 710 were enrolled and randomly assigned in a 1:1 ratio to either the Vitamin C group (n = 355) or the control group (n = 355). The majority of exclusions during screening were due to failure to meet the inclusion criteria.

All participants received intramuscular administration of 2 units of hemocoagulase bothrops atrox before the biopsy. Participants allocated to the Vitamin C group were scheduled to receive two intravenous infusions of vitamin C (20 mg/kg each): the first dose 6 h before the biopsy and the second dose 18 h after the biopsy. Participants in the control group received no additional prophylactic hemostatic agents and no cross-over between groups occurred.

A protocol deviation occurred in five participants (1.41%) in the Vitamin C group, who did not receive the second postoperative dose due to missing the predefined administration window after leaving the study unit for required diagnostic tests. All patients completed postoperative renal ultrasound examinations and the Pain VAS self-assessment as scheduled. No participants were lost to follow-up for the primary and secondary outcomes. All 710 participants were included in the ITT analysis. A summary of patient disposition is presented in Fig. 1.

Demographics and baseline characteristics

Baseline characteristics were presented in Table 1 and Additional file 1: Table S1. The mean age was 46.15 ± 14.67 years in the Vitamin C group and 49.25 ± 14.25 years in the control group ($P = 0.004$). All other demographic, laboratory, and clinical parameters, medical histories, and the use of concomitant medications were similarly distributed. Although age differed statistically between groups, the effect sizes were small (standardized mean difference: 0.214), suggesting limited clinical relevance. Overall, baseline characteristics were well-matched, supporting the comparability of the two groups for subsequent efficacy analyses.

Biopsy procedure and pathological diagnoses

Biopsy procedural details and the spectrum of pathological diagnoses were balanced between the two groups

Enrolment

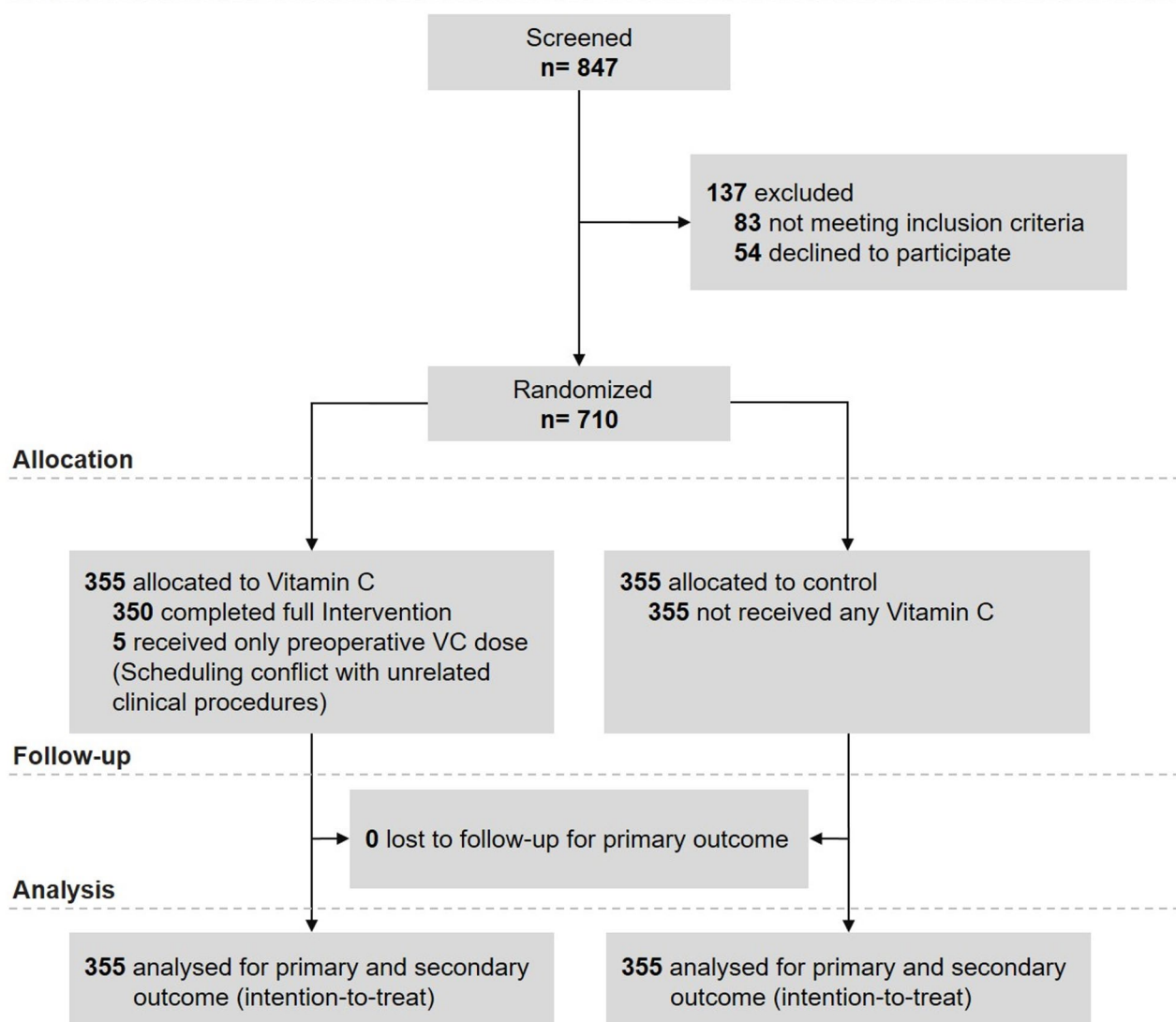


Fig. 1 Study patient disposition

(Table 2 and Additional file 1: Table S2). The majority of biopsies required two needle insertions (Vitamin C: 98.6%; Control: 97.2%), and all but two procedures (both in the control group) were performed on the right kidney. The most common pathological diagnosis in both groups was IgA nephropathy (31.3% each), followed by membranous nephropathy (Vitamin C: 19.4%; Control: 24.2%). The distribution of all other diagnostic categories was similar between groups (all $P > 0.05$).

Outcomes

Data for the primary and secondary outcomes were available for all 710 randomized participants.

Primary outcome

Perirenal hematoma area

In Table 3, the unadjusted mean difference in hematoma area at 24 h was 1.65 mm^2 (95% CI, -54.764 to 58.054 ; $P = 0.954$). After adjustment for baseline age imbalance using ANCOVA, the adjusted mean difference was 4.44 mm^2 (95% CI, -52.31 to 61.18 mm^2 ; $P = 0.878$). The model explained minimal variance ($R^2 < 0.001$).

Pain VAS scores

At all time points, the median VAS score in both groups was 0. The raw P value suggested a difference at 2 h post-biopsy ($P = 0.035$), whereas the differences at 12 and 24 h were not significant ($P = 0.064$ and $P = 0.178$, respectively) (Additional file 1: Table S3). After Bonferroni correction for three tests, none of the comparisons remained

Table 1 Demographics and baseline characteristics

Characteristic	Vitamin C N = 355	Control N = 355
Age at screening, yr	46.15 ± 14.67	49.25 ± 14.25
Female sex, n (%)	146 (41.1)	145 (40.8)
BMI, kg/m ²	24.81 ± 3.99	24.67 ± 3.75
Blood pressure, mm Hg		
Systolic	126.30 ± 14.72	125.92 ± 16.26
Diastolic	79.91 ± 9.57	80.75 ± 10.18
Albumin, g/dl	3.91 [3.30, 4.40]	3.89 [3.06, 4.44]
Urea, mg/dl	33.24 [24.76, 49.28]	35.58 [27.35, 52.41]
Creatinine, mg/dl	0.92 [0.69, 1.50]	0.93 [0.71, 1.52]
Uric acid, mg/dl	6.05 ± 1.93	6.14 ± 1.96
Cystatin C, mg/dl	0.12 [0.09, 0.17]	0.12 [0.09, 0.17]
eGFR, mL/min per 1.73m ²	94.18 [51.83, 116.05]	87.37 [47.72, 109.23]
Cholesterol, mg/dl	182.52 [147.72, 227.77]	189.87 [153.91, 237.43]
Triglycerides, mg/dl	165.63 [110.71, 224.45]	150.57 [111.60, 224.08]
Glucose, mg/dl	52.3 [47.7, 59.8]	52.8 [47.6, 58.2]
24-hour urine protein, g	1.49 [0.76, 3.37]	1.71 [0.78, 4.21]
UACR, g/g	1.02 [0.39, 2.81]	1.37 [0.41, 3.64]
PT, s	8.76 ± 0.58	8.77 ± 0.62
APTT, s	31.32 ± 3.79	31.80 ± 4.19
TT, s	18.11 ± 1.22	18.11 ± 1.30
INR	0.97 ± 0.06	0.97 ± 0.07
Medical History, n (%)		
Hypertension	160 (45.1)	182 (51.3)
Diabetes mellitus	66 (18.6)	66 (18.6)
CVD	33 (9.3)	32 (9.0)
Concomitant Medications, n (%)		
Antihypertensive agents	271 (76.3)	280 (78.9)
Glucose-lowering agents	113 (31.8)	94 (26.5)
Lipid-modifying agents	156 (43.9)	146 (41.1)
Immunosuppressants	38 (10.7)	42 (11.8)
Anemia & Bone Mineral Metabolism	71 (20.0)	72 (20.3)

Continuous data are presented as mean ± SD (normally distributed) or median [IQR] (non-normally distributed). Categorical data are presented as n (%). P values for continuous variables are from t-tests or Mann-Whitney U tests, and from χ^2 or Fisher's exact tests for categorical variables

BMI, body mass index; UACR, urine albumin-to-creatinine ratio; PT, prothrombin time; APTT, activated partial thromboplastin time; TT, thrombin time; INR, international normalized ratio; CVD, cardiovascular disease

statistically significant (adjusted $P=0.105$, 0.192 , and 0.534 , respectively) (Table 3).

A zero-inflated negative binomial regression was performed to account for the substantial proportion of zero scores and to adjust for baseline age (Additional file 1: Table S4 and Figure S4). However, age was a significant influencing factor. Older age was not only associated with a lower expected pain intensity (at 12 h: rate ratio [RR] = 0.984 , $P=0.001$) but also with reduced odds of reporting zero pain (at 2 h: odds ratio [OR] = 1.019 , $P<0.001$).

Secondary outcomes

Bleeding events and complications

There were no significant differences between groups in the incidence of perirenal hematoma or other bleeding complications (Table 4). The proportion of participants

with any perirenal hematoma was 56.3% in the vitamin C group and 50.1% in the control group (RR, 1.124 ; 95% CI, 0.978 – 1.290 ; $P=0.098$). There was no statistically significant difference in the distribution of perirenal hematoma severity between the Vitamin C and Control groups ($\chi^2 = 1.542$, $P=0.214$). The rates of gross hematuria and transfusion requirement were also comparable between groups. No patient in either group required Surgical or interventional surgery for bleeding.

Postbiopsy pain severity trends

The distribution of local pain severity at 2, 12, and 24 h postoperatively is detailed in Table 4. There was a statistically significant difference in the pain severity distribution at 12 h (P for trend = 0.033), favoring the control group which had a higher proportion of pain-free

Table 2 Biopsy information and pathological diagnoses

Characteristic, n (%)	Vitamin C N = 355	Control N = 355	P value
Needle insertions			
2	350 (98.6)	345 (97.2)	0.192
3	5 (1.4)	10 (2.8)	
Left renal biopsy	0	2 (0.6)	0.499
Primary Renal Diagnosis^a			
IgA nephropathy	111 (31.3)	103 (31.3)	0.513
Membranous nephropathy	69 (19.4)	86 (24.2)	0.122
Diabetic nephropathy	47 (13.2)	38 (10.7)	0.298
Hypertensive renal disease ^b	34 (9.6)	21 (5.9)	0.068
Minimal change disease	15 (4.2)	17 (4.8)	0.718
Focal segmental glomerulosclerosis	12 (3.4)	15 (4.2)	0.556
Lupus nephritis	6 (1.7)	8 (2.3)	0.589
Other glomerulonephritides ^c	47 (13.2)	51 (14.4)	0.663

^aIn cases of multiple diagnostic entries, only the highest-priority diagnosis was selected for classification

^bHypertensive renal disease was classified as such only when it was the primary or sole diagnosis

^cIncludes all remaining diagnostic categories with individually low frequencies. Full distribution provided in Supplementary Table 2

Table 3 Primary Outcomes

Analysis model	Vitamin C group N = 355	Control group N = 355	Effect size (95% CI)	P value
Perirenal hematoma				
All Participants				
Any hematoma, n (%)	200 (56.3)	178 (50.1)	RR:1.124 (0.978, 1.290)	0.098
Crude area, mm ²	191.02 ± 274.89	189.37 ± 466.36	MD:1.645 (−54.764, 58.054)	0.954
Adjusted mean area ^a , mm ²	192.42	187.97	MD:4.435 (−52.305, 61.176)	0.878
Participants with Hematoma				
Area, mm ²	339.06 ± 289.78	377.68 ± 602.87	MD:−38.625 (−132.690, 55.440)	0.420
Pain VAS scores				
2 h ^b	0 [0, 9]	0 [0, 6]	–	0.105
12 h ^b	0 [0, 7]	0 [0, 5]	–	0.192
24 h ^b	0 [0, 4]	0 [0, 3]	–	0.534

^aAdjusted for baseline age by ANCOVA. Values presented are estimated marginal means. Effect size is mean difference (MD) or risk ratio (RR)

^bData are presented as median [IQR] and were compared using the Mann-Whitney U test. Adjusted P-values were calculated using the Bonferroni correction for comparisons across 3 time points

participants (74.6% vs. 67.3%). At 2 h, a non-significant trend was observed ($P=0.095$). By 24 h, the distribution between groups was similar, with no evidence of a linear trend ($P=0.459$).

Changes in laboratory and vital sign parameters

Postbiopsy urinary vitamin C concentration increased significantly in the vitamin C group compared with the control group (Diff, 25.67 ± 18.09 mg/L vs. -1.01 ± 9.66 mg/L; SMD, 1.839; 95% CI, 1.663 to 2.014; $P<0.001$), confirming adherence to the protocol. No significant between-group differences were observed in the postbiopsy changes from baseline in systolic blood pressure, hemoglobin level, or platelet count (Table 5).

Subgroup analyses

In post-hoc exploratory subgroup analyses (Additional file 1: Table S7–S9), we observed no significant interaction between treatment and baseline urinary vitamin C levels for the primary outcome of hematoma area (tertiles $p=0.422$; quartiles $p=0.484$) or for pain outcomes (all p for interaction >0.05). No subgroup benefited from vitamin C.

Adverse events

Adverse events were monitored from the first administration of the intervention until the end of study. The overall incidence of AEs was low and comparable between the two groups: 4.2% (15 of 355) in the Vitamin C group and 3.7% (13 of 355) in the control group (Table 6 and Additional file 1: Table S5). Gastrointestinal disorders were the most frequently reported events. All reported AEs

Table 4 Secondary outcomes

Outcome, n (%)	Vitamin C N = 355	Control N = 355	Risk Ratio (95% CI)	P value ^a
Perirenal hematoma severity				
Mild (≤ 200 mm ²)	71 (20.0)	61 (17.2)	1.164 (0.855, 1.585)	0.335
Moderate (201–400 mm ²)	76 (21.4)	67 (18.9)	1.134 (0.846, 1.522)	0.400
Severe (> 400 mm ²)	53 (14.9)	50 (14.1)	1.060 (0.742, 1.515)	0.749
P for trend ^b			$\chi^2 = 1.542$	0.214
Gross hematuria	7 (2.0)	2 (0.6)	3.500 (0.732, 16.732)	0.093
Transfusion required	0	2 (0.6)	–	0.499
Intervention for bleeding	0	0	–	–
Pain Severity over time				
2 h				
Pain-free (0–4)	238 (67.0)	260 (73.2)	0.915 (0.831, 1.008)	0.071
Mild (5–44)	98 (27.6)	81 (22.8)	1.210 (0.938, 1.561)	0.142
Moderate (45–74)	15 (4.2)	10 (2.8)	1.500 (0.683, 3.294)	0.309
Severe (> 75)	4 (1.1)	4 (1.1)	–	–
P for trend ^b			$\chi^2 = 2.791$	0.095
12 h				
Pain-free (0–4)	239 (67.3)	265 (74.6)	0.902 (0.821, 0.991)	0.032
Mild (5–44)	95 (26.8)	77 (21.7)	1.234 (0.949, 1.603)	0.115
Moderate (45–74)	18 (5.1)	10 (2.8)	1.800 (0.843, 3.845)	0.123
Severe (> 75)	3 (0.8)	3 (0.8)	–	–
P for trend ^b			$\chi^2 = 4.567$	0.033
24 h				
Pain-free (0–4)	277 (78.0)	285 (80.3)	0.972 (0.901, 1.048)	0.460
Mild (5–44)	72 (20.3)	66 (18.6)	1.091 (0.808, 1.472)	0.569
Moderate (45–74)	6 (1.7)	3 (0.8)	2.000 (0.504, 7.935)	0.505
Severe (> 75)	0	1 (0.3)	–	–
P for trend ^b			$\chi^2 = 0.549$	0.459

^aP value for between-group comparison, χ^2 or Fisher's exact test^bCochran-Mantel-Haenszel test for linear trend ($\chi^2 = 1.542$, df = 1)**Table 5** Changes in post-laboratory and vital sign parameters

Parameter	Time Point	Vitamin C N = 355	Control N = 355	Between-Group Difference SMD (95% CI)	P value
Urinary Vitamin C, mg/l	Baseline	32.10 $\pm$ 13.11	32.68 $\pm$ 16.15	–0.040 (–0.187, 0.107)	0.596
	Postbiopsy	57.76 $\pm$ 25.00	31.67 $\pm$ 16.67	1.228 (1.067, 1.388)	<0.001
	Diff.	25.67 $\pm$ 18.09	–1.01 $\pm$ 9.66	1.839 (1.663, 2.014)	<0.001
Systolic BP, mm Hg	Baseline	126.30 $\pm$ 14.72	125.92 $\pm$ 16.26	0.024 (–0.123, 0.171)	0.749
	Postbiopsy	123.48 $\pm$ 16.26	124.48 $\pm$ 17.23	–0.060 (–0.207, 0.087)	0.427
	Diff.	–2.81 $\pm$ 12.80	–1.44 $\pm$ 14.72	–0.100 (–0.247, 0.048)	0.185
Hemoglobin, g/dl	Baseline	13.50 $\pm$ 2.47	13.36 $\pm$ 2.52	0.054 (–0.093, 0.201)	0.469
	Postbiopsy	13.16 $\pm$ 2.27	12.96 $\pm$ 2.41	0.086 (–0.061, 0.233)	0.251
	Diff.	–0.34 $\pm$ 1.10	–0.40 $\pm$ 1.07	0.061 (–0.087, 0.208)	0.419
Platelet, 10 ⁹ /l	Baseline	227.20 $\pm$ 71.91	228.11 $\pm$ 74.24	–0.012 (–0.160, 0.135)	0.869
	Postbiopsy	222.03 $\pm$ 68.37	221.89 $\pm$ 72.58	0.002 (–0.145, 0.149)	0.978
	Diff.	–5.17 $\pm$ 47.17	–6.22 $\pm$ 48.15	0.022 (–0.125, 0.169)	0.769

were Grade 1 or 2 (Additional file 1: Table S6). No serious AEs were reported, and no specific AEs associated with Vitamin C, such as hemolysis, nephrolithiasis, or allergic reactions, were observed.

Discussion

This was a prospective, open-label, randomized controlled trial designed to evaluate the efficacy and safety of intravenous vitamin C for preventing bleeding and pain complications after PRB in CKD patients. The intervention regimen was based on the pharmacokinetic profile

Table 6 Adverse events during the study period

Preferred Term, n (%)	Vitamin C N=355	Control N=355
Any adverse events	15 (4.23)	13 (3.66)
Gastrointestinal Disorders	9 (2.54)	5 (1.41)
General Conditions and Administration Site Conditions	2 (0.56)	2 (0.56)
Respiratory, Thoracic and Mediastinal Disorders	0	1 (0.28)
Musculoskeletal and Connective Tissue Disorders	3 (0.85)	1 (0.28)
Nervous System Disorders	3 (0.85)	3 (0.85)
Renal and Urinary Disorders	0	2 (0.56)
Psychiatric Disorders	0	1 (0.28)

AE grading followed CTCAE v5.0. Incidence (%) = (n/N) × 100, where n is the number of patients with the AE

of vitamin C, aiming to maintain effective plasma concentrations. The primary results demonstrated that vitamin C failed to reduce the perirenal hematoma area at 24h post-biopsy or alleviate pain at puncture site within 24h.

Our study revealed that, compared with control, vitamin C was not effective in preventing perirenal hematoma after kidney biopsy. The between-group difference was not statistically significant and was clinically negligible. The analysis of pain outcomes presented complexity. No significant difference was found based on the continuous VAS scores after multiple testing correction. However, an exploratory analysis based on ordered categories suggested a trend toward less favorable pain severity distribution in the vitamin C group at 12h. Regarding safety, all serious adverse events were unrelated to the study intervention, indicating the overall safety of the dosing regimen.

The negative outcomes for both primary endpoints revealed that vitamin C did not provide clinical benefits to CKD patients with PRB in preventing post-biopsy complications. Several factors may explain the gap between these theoretical benefits and our negative findings. While vitamin C can contribute to vascular repair by promoting collagen synthesis [19, 20], or to postoperative oxidative stress and inflammation by inhibiting the NF-κB pathway [21], these are chronic, adaptive processes. In contrast, post-biopsy bleeding and pain are immediate consequences of acute mechanical trauma [22], involving direct vascular injury and a rapid local inflammatory response. Furthermore, the administered dose may have been inadequate to achieve effective tissue concentrations for acute hemostasis or analgesia. The characteristically rapid metabolism and renal clearance of vitamin C could have further limited the efficacy of our short-term perioperative regimen, preventing the maintenance of sufficient concentration during the window following injury. This pharmacokinetic challenge

is supported by our observed changes in urinary vitamin C levels before and after the procedure. Thus, while chronic supplementation can improve conditions like gingival bleeding [23], temporary administration of vitamin C during the perioperative period of biopsy was not effective.

Although an exploratory analysis suggested a less favorable distribution of pain severity in the vitamin C group at 12h, a more robust zero-inflated negative binomial regression model [24], after adjusting for the strong confounder of age, showed no significant association between vitamin C and pain intensity at any time point. This inconsistency may be influenced by changes in pain sensitivity with aging [25]. Generally, our data still do not support a pain-relieving benefit for vitamin C post-biopsy, after effectively eliminating the influence of unbalanced age on pain reporting.

Our study adhered rigorously to a prospective randomized controlled design, providing a high-quality framework. The pharmacokinetic-based intervention regimen enhanced the rigor of the hypothesis testing. Blinded assessment of the primary outcome, with quantitative measurements performed by radiologists unaware of group allocation, significantly minimized measurement bias for the primary endpoint. All randomized participants were included in the ITT analysis, adhering to the study design principles. However, this study has several limitations. The open-label design may have introduced reporting bias. Pain-related analyses across multiple time points were exploratory; although the primary outcome results are robust, findings on pain should be considered hypothesis-generating. Only one dose and schedule of vitamin C was tested. Hematoma area measurement is operator-dependent and pain VAS is subjective, despite our use of blinded assessors. Exclusion of high-risk patients limits generalizability to those with bleeding diatheses, uncontrolled hypertension, or structural kidney abnormalities. Urinary vitamin C concentration does not reliably reflect systemic vitamin C status; although a high urinary level suggests relative sufficiency in some patients, a low level does not necessarily indicate deficiency, and the absence of plasma vitamin C data is a limitation.

Our results contrast with those of earlier, sample-limited interventional studies [13, 14, 16, 26]. This discrepancy may stem from differences in participant populations, the timing and dosing of the intervention or study design. Our findings showed that a short-term, temporary use of vitamin C to prevent bleeding and pain is not recommended for CKD patients undergoing PRB. Given the negative results of this large trial, future research may focus on alternative dosing strategies or higher-risk populations such as vitamin C-deficient patients.

Conclusions

Our study demonstrates that intravenous vitamin C does not prevent bleeding or alleviate pain after kidney biopsy in CKD patients and therefore should not be used routinely in this clinical procedure.

Abbreviations

AE	Adverse Event
ANCOVA	Analysis of Covariance
APTT	Activated Partial Thromboplastin Time
BMI	Body Mass Index
BP	Blood Pressure
CI	Confidence Interval
CKD	Chronic Kidney Disease
CMH	Cochran-Mantel-Haenszel
CTCAE	Common Terminology Criteria for Adverse Events
CVD	Cardiovascular Disease
eGFR	estimated Glomerular Filtration Rate
INR	International Normalized Ratio
ITT	Intention-to-Treat
MD	Mean Difference
OR	Odds Ratio
PRB	Percutaneous Renal Biopsy
PT	Prothrombin Time
RR	Risk Ratio
SMD	Standardized Mean Difference
TT	Thrombin Time
UACR	Urine Albumin-to-Creatinine Ratio
VAS	Visual Analogue Scale

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12916-026-05037-x>.

Supplementary Material 1: Additional File 1: Supplementary Methods, Results, and Definitions. This file contains: Supplementary Methods (post-hoc subgroup analysis), Supplementary Results (Tables S1–S9, Figure S1), and Supplementary Definitions (medical history and biopsy procedure).

Supplementary Material 2: Additional file 2: The full study protocol (v2.0).

Acknowledgements

The authors would like to thank all participating patients, the inpatient department staff, the interventional ultrasound department, and the medical statistics teaching and research office at our institution. A senior professor assisted in the study design and statistical analysis.

Author contributions

The co-first authors contributed equally to this work. They had full access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Concept and design: co-first authors and corresponding author. Investigation: all investigators. Statistical analysis: co-first authors. Drafting of the manuscript: co-first authors. Critical revision: all authors. Administrative, technical, or material support: all investigators. Supervision: corresponding author. All authors read and approved the final manuscript.

Funding

This study was in part by grants from the affiliated hospital of a national medical university.

Data availability

A deidentified participant data that support the findings of this study are available upon request from the corresponding author upon reasonable request, or can access whole data from the Clinical Trial Management Public Platform (ResMan, <http://www.medresman.org.cn>) within 6 months after posting the result.

Declarations

Ethics approval and consent to participate

Ethics approval was obtained from the institutional ethics committee. Informed consent was obtained from all individual participants included in the study. The trial has been registered with the Chinese Clinical Trial Registry (ChiCTR2500106962).

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 22 February 2026 / Accepted: 25 June 2026

Published online: 03 July 2026

References

1. Poggio ED, McClelland RL, Blank KN, Hansen S, Bansal S, Bombardier AS, et al. Systematic review and meta-analysis of native kidney biopsy complications. *Clin J Am Soc Nephrol*. 2020;15:1595–602.
2. Tondel C, Vikse BE, Bostad L, Svarstad E. Safety and complications of percutaneous kidney biopsies in 715 children and 8573 adults in Norway 1988–2010. *Clin J Am Soc Nephrol*. 2012;7:1591–7.
3. Lees JS, McQuarrie EP, Mordi N, Geddes CC, Fox JG, Mackinnon B. Risk factors for bleeding complications after nephrologist-performed native renal biopsy. *Clin Kidney J*. 2017;10:573–7.
4. Xu DM, Chen M, Zhou FD, Zhao MH. Risk factors for severe bleeding complications in percutaneous renal biopsy. *Am J Med Sci*. 2017;353:230–5.
5. Manno C, Bonifati C, Torres DD, Campobasso N, Schena FP. Desmopressin acetate in percutaneous ultrasound-guided kidney biopsy: a randomized controlled trial. *Am J Kidney Dis*. 2011;57:850–5.
6. Wang C, Lebedeva V, Yang J, Anih J, Park LJ, Paczkowski F, et al. Desmopressin to reduce periprocedural bleeding and transfusion: a systematic review and meta-analysis. *Perioper Med (Lond)*. 2024;13:5.
7. Mohinani A, Patel S, Tan V, Kartika T, Olson S, DeLoughery TG, et al. Desmopressin as a hemostatic and blood sparing agent in bleeding disorders. *Eur J Haematol*. 2023;110:470–9.
8. Lim CC, Siow B, Choo JCI, Chawla M, Chin YM, Kee T, et al. Desmopressin for the prevention of bleeding in percutaneous kidney biopsy: efficacy and hyponatremia. *Int Urol Nephrol*. 2019;51:995–1004.
9. Izawa J, Matsuzaki K, Raita Y, Uehara G, Nishioka N, Yano H, et al. Intravenous tranexamic acid in percutaneous kidney biopsy: a randomized controlled trial. *Nephron*. 2023;147:144–51.
10. May JM, Harrison FE. Role of vitamin C in the function of the vascular endothelium. *Antioxid Redox Signal*. 2013;19:2068–83.
11. Kwon JH, Kim D, Cho H, Shin BS. Ascorbic acid improves thrombotic function of platelets during living donor liver transplantation by modulating the function of the E3 ubiquitin ligases c-Cbl and Cbl-b. *J Int Med Res*. 2019;47:1856–67.
12. Gęgotek A, Skrzydlewska E. Antioxidative and anti-inflammatory activity of ascorbic acid. *Antioxidants*. 2022;11(10):1993.
13. Farahani-Jam S, Yaghmaei M, Vahdat Shariatpanahi Z. Effect of intravenous ascorbic acid administration on hemorrhage and wound complications in total abdominal hysterectomy: A prospective randomized clinical trial. *Clin Nutr ESPEN*. 2022;49:74–78.
14. Vahdat Shariatpanahi Z, Shahbazi S, Shahbazi E. Ascorbic acid to prevent postpolypectomy bleeding in the colon: a randomized controlled trial. *Asian J Endosc Surg*. 2022;15:103–9.
15. Hung KC, Lin YT, Chen KH, Wang LK, Chen JY, Chang YJ, et al. The effect of perioperative vitamin C on postoperative analgesic consumption: a meta-analysis of randomized controlled trials. *Nutrients*. 2020;12(10):3109.
16. Rachana ND, Ranganath N, Pallavi VR, Shashidhar GS, Arathi BH, Gowda VB. Effect of intravenous vitamin C and N acetylcysteine on postoperative pain and opioid consumption after laparoscopic gynaecologic oncosurgeries: a randomised controlled study. *J Clin Diagn Res*. 2024;18(7):UC07–UC11.
17. Padayatty SJ, Sun H, Wang Y, Riordan HD, Hewitt SM, Katz A, et al. Vitamin C pharmacokinetics: implications for oral and intravenous use. *Ann Intern Med*. 2004;140:533–7.

18. Institutions MUBoSafN-gM. Practice guidelines for percutaneous ultrasound-guided renal biopsy. *Chin J Med Ultrasound (Electronic Edition)*. 2021;18:1023–43.
19. Aguirre R, May JM. Inflammation in the vascular bed: importance of vitamin C. *Pharmacol Ther*. 2008;119:96–103.
20. Calvino N. Connective tissue: Vascular and hematological (blood) support. *J Chiropr Med*. 2003;2:25–36.
21. Kietzmann T. Vitamin C: From nutrition to oxygen sensing and epigenetics. *Redox Biol*. 2023;63:102753.
22. Kennedy SA, Milovanovic L, Midia M. Major bleeding after percutaneous image-guided biopsies: frequency, predictors, and periprocedural management. *Semin Intervent Radiol*. 2015;32:26–33.
23. Tada A, Miura H. The relationship between vitamin C and periodontal diseases: a systematic review. *Int J Environ Res Public Health*. 2019;16(14):2472.
24. Wang Z, Ling W, Wang T. A semiparametric quantile regression rank score test for zero-inflated data. *Biometrics*. 2025;81(2):ujaf050.
25. Lautenbacher S, Peters JH, Heesen M, Scheel J, Kunz M. Age changes in pain perception: A systematic-review and meta-analysis of age effects on pain and tolerance thresholds. *Neurosci Biobehav Rev*. 2017;75:104–13.
26. Jeon Y, Park JS, Moon S, Yeo J. Effect of intravenous high dose vitamin C on postoperative pain and morphine use after laparoscopic colectomy: a randomized controlled trial. *Pain Res Manag*. 2016;2016:9147279.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.